Diagnostic Report generated by gemini-2.5-pro

Diagnostic Analysis of Run_25

1. Summary of Key Deviations

The analysis of Run_25 indicates a significant over-performance of the fuel delivery system, coupled with a persistent sensor failure.

- **Flow_Pump (Moderate):** The most significant physical anomaly is a consistent positive **offset and drift**, with flow reaching **+12.2%** higher than the healthy baseline. This suggests the pump is delivering substantially more fuel than expected.
- **Flow_Bypass (Moderate):** This signal mirrors the pump's behavior, showing a strong positive **offset and drift** up to **+14.5%**. This is a direct physical consequence of the increased pump flow being diverted.
- **Flow_Engine1-4 (Minor):** All four engine nozzles show a consistent minor positive **offset** (up to **+6.7%**) during the high-throttle phases, indicating more fuel is reaching the engine than normal.
- **System Pressures (p_FMU, p_Shut) (Minor):** Interestingly, there is no corresponding increase in system pressure. In fact, `p_FMU` and `p_Shut` show a slight **negative offset** (approx. -3%) in later states.
- **Flow_PRV (Critical):** The signal remains in a state of **critical saturation** (+1067% to +11250%) across all states, confirming the ongoing, isolated failure of this sensor.

2. Root-Cause Hypotheses

Rank	Suspected Component	Fault Type	Supporting Evidence
1	**Main Fuel Pump**	**Significant Over-performance / Degradation**	The pump is delivering ~12% more flow than commanded. This is the primary physical deviation and directly causes the increased flow seen in the **Flow_Bypass** and **Flow_Engine** sensors. This type of failure could result from internal wear or a control malfunction causing the pump to operate at a higher effective displacement.

2	**Fuel Metering Unit (FMU)**	**Stuck Partially Open / Calibration Shift**	If the FMU were allowing more fuel through than intended for a given throttle position, it could lead to increased **Flow_Engine**. However, this wouldn't explain the high **Flow_Pump** reading. The pump over-delivery is the more likely upstream cause.
3	**Bypass Valve**	**Stuck Partially Closed / High Cracking Pressure**	If the bypass valve were restricted, it would force more flow towards the engine and PRV, increasing **Flow_Engine** and potentially **Flow_Pump** readings. This would also likely cause a significant *increase* in system pressure, which is not observed here, making this hypothesis less likely.
4	**p_FMU / p_Shut Sensors**	**Minor Calibration Error (Negative Offset)**	The fact that pressures are slightly *low* while flows are high is physically inconsistent. It's plausible that these pressure sensors have a slight negative calibration bias, and the real system pressure is actually slightly elevated due to the high flow.
5	**Flow_PRV Sensor**	**Sensor Saturation / Failure**	This is a confirmed, but separate, electrical fault. It is a "red herring" that does not contribute to the primary physical issue of fuel over-delivery.

3. Most Probable Hypothesis

The most probable root cause is a **Significant Over-performance of the Main Fuel Pump**, likely due to internal degradation or a control-side fault.

Reasoning: The evidence points squarely to the pump as the origin of the fault. The **+12.2%** positive offset in `Flow_Pump` is the largest and most persistent *physical* deviation in the system. This single upstream fault provides a direct and sufficient physical explanation for the secondary effects seen downstream: increased bypass flow (`Flow_Bypass` at +14.5%) and increased engine flow (`Flow_Engine1-4` at +6.7%). The lack of a corresponding pressure *increase* is puzzling but could be explained by a minor, secondary calibration error on the pressure sensors or by the system's ability to absorb the extra flow. The pump over-delivery is the most comprehensive explanation for the majority of the observed data.